SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL

1. Exercise

a) We denote states of the buffer i as 0, 1 and 2, respectively, based on the number of pieces of type i available. The event set is $E = \{a_i, p\}$ and the associated transition diagram is the usual transition diagram of a finite queue, as shown in Fig. 1.1. [4]

b) The assembly robot operates with two buffers, and it can be modeled through the automaton $G = G_1||G_2$, shown in Fig. 1.2. [4]

c) The non-deterministic automaton G_N is obtained from G simply replacing events a_1 and a_2 by events a. This is shown in Fig. 1.3. [4]

d) The observer automaton $\text{Obs}(G_N)$ is shown in Fig. 1.4; This automaton keeps track of the possible occupancies of individual buffers 1 and 2 following any word $w \in \{a, p\}^*$. [6]

e) In particular, $f_{obs}(\{00\}, aaaad) = \{11\}$. This means that both buffers have a single piece (of type 1 and 2 respectively). Instead: $f_{obs}(\{00\}, aaada) = \{02, 11, 20\}$ this means that in total two pieces are in the buffers, but we do not know the type of the pieces available. [2]

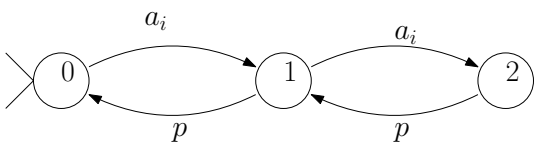


Figure 1.1 Automaton G_i, modelling buffer i occupancy

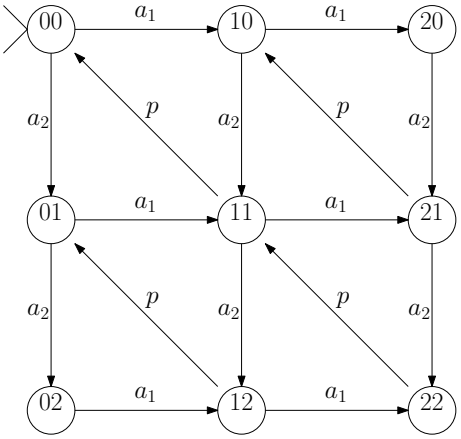


Figure 1.2 Parallel composition $G = G_1||G_2$

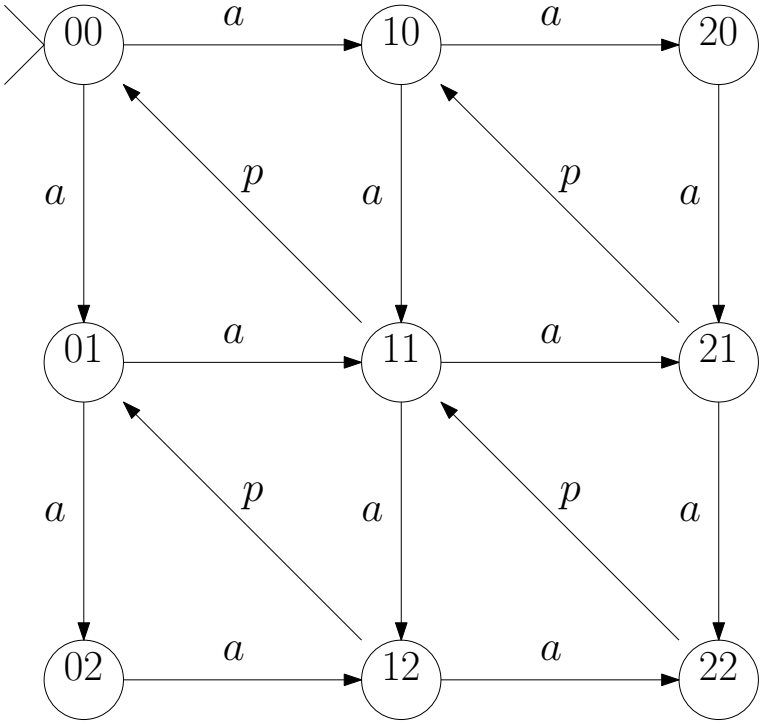


Figure 1.3 Non deterministic automaton G_N

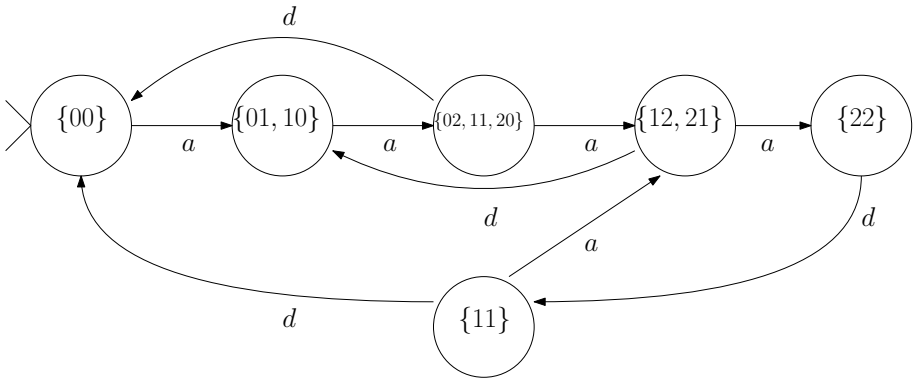


Figure 1.4 Observer automaton Obs(G_N)

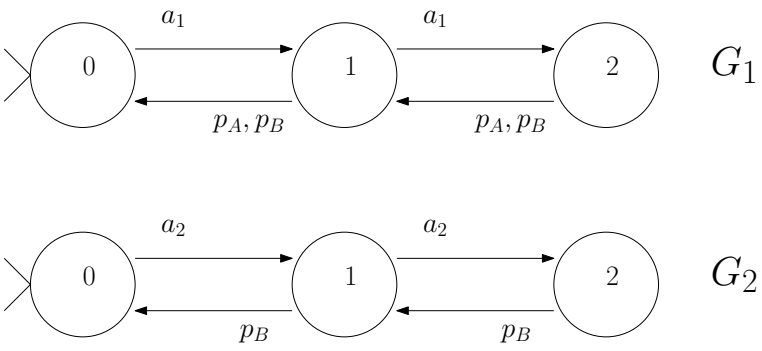


Figure 2.1 Transition diagrams of G_1 and G_2

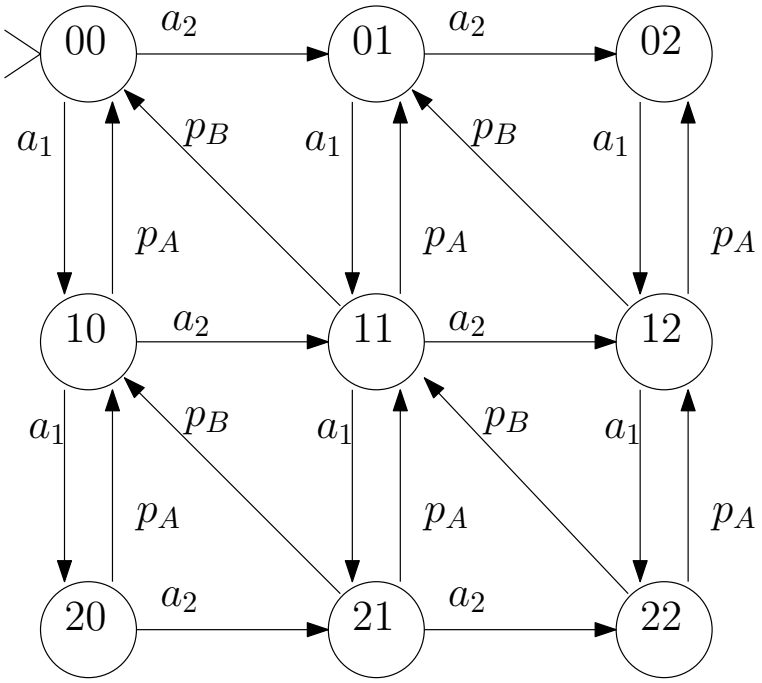


Figure 2.2 Transition diagrams of $G_1||G_2$

2. Exercise

a) Modelling the Automata G_1 and G_2 can be accomplished as in Fig. 2.1. [3]

b) The parallel composition $G = G_1||G_2$ is shown in Fig. 2.2. [3]

c) The partial specification automaton G_K is shown in Fig. 2.3. [3]

d) The concurrent composition $G||G_K$ allows to check if the specification is controllable. Notice in particular that no a_1 or a_2 events are ever disabled in $G||G_K$

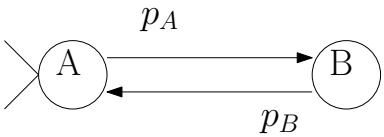


Figure 2.3 Transition diagrams of G_K

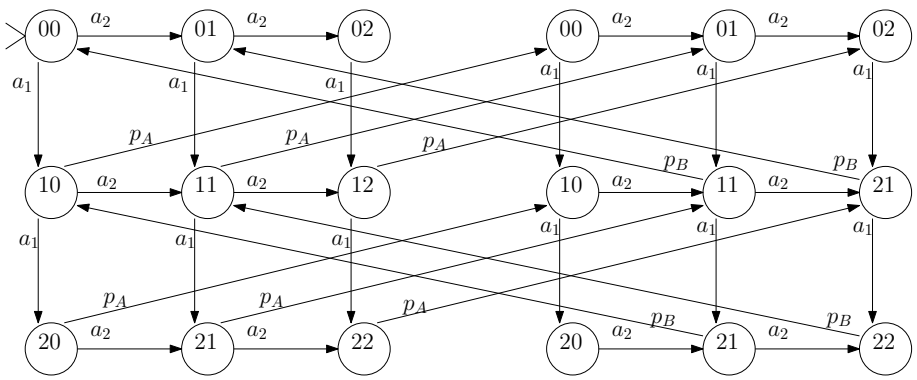


Figure 2.4 Transition diagram of $G||G_K$

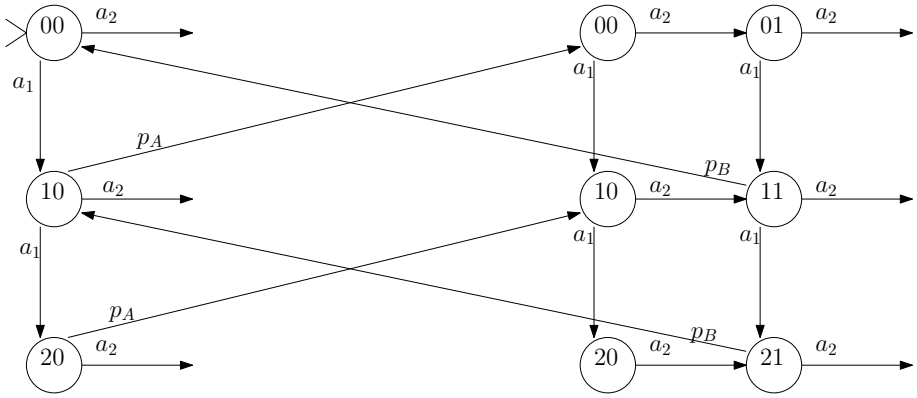


Figure 2.5 Supervisor automaton

(viz. $[\Gamma_G(x) \backslash \Gamma_{G||G_K}(x,x_K)] \cap \{a_1,a_2\} = \emptyset$ for all $x \in X_G$ and all $x_K \in \{A,B\}$). Therefore the specification is controllable. See Fig. 2.4. [4]

e) It can be seen in Fig. 2.4 that event p_B is disabled in states $11A, 12A, 21A, 22A$. Hence the specification is uncontrollable. [2]

f) The automaton in Fig. 2.5 represents an admissible supervisor S which implements the maximal controllable sublanguage. [5]

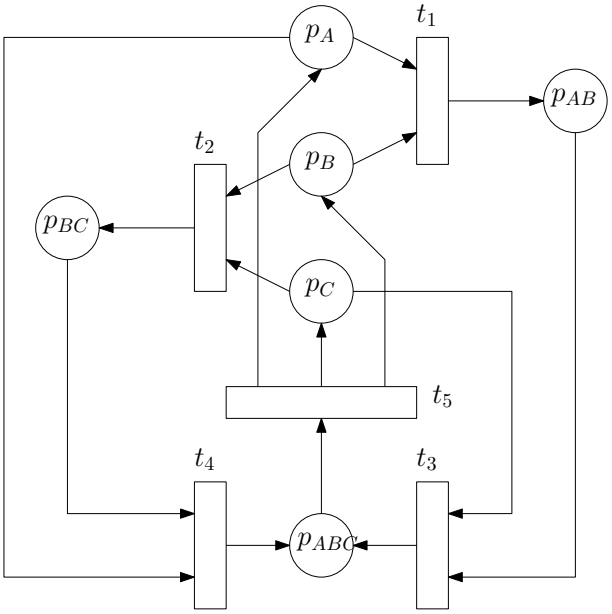


Figure 3.1 Petri Net N

3. Exercise

a) The chemical reaction network can be modeled by the Petri Net N sketched in Fig. 3.1. [4]

b) Listing species in the following order $p_A, p_B, p_C, p_{AB}, p_{BC}, p_{ABC}$ the incidence matrix is given by:

$$C = \begin{bmatrix} -1 & 0 & 0 & -1 & 1 \\ -1 & -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 \end{bmatrix}$$

[2]

c) The P semiflows of minimal support correspond to the conservation of molecules of A, B and C, respectively. In particular, these are given by:

$$[1,0,0,1,0,1], \quad [0,1,0,1,1,1], \quad [0,0,1,0,1,1];$$

[4]

d) The T semiflows are given by: $[1,0,1,0,1]', [0,1,0,1,1]'$ [4]

e) The transition diagram associated to the reachable set is shown in Fig. 3.2. [3]

f) The marked Petri Net (N, M_0) is not reversible. Notice that marking $[0,0,0,1,1,0]$ is the unique absorbing state, and therefore with probability 1 all evolutions will converge to it. Therefore we expect to find 1 molecule of AB and one of BC for sufficiently large times. [3]

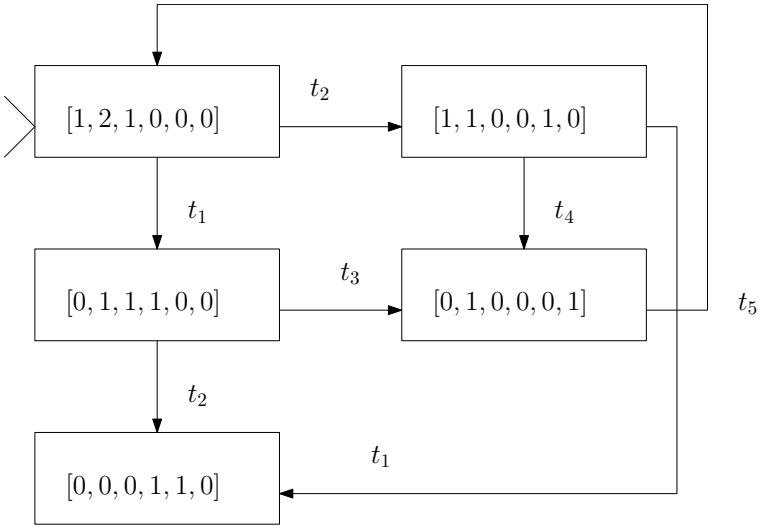


Figure 3.2 Reachable set of Petri Net (N, M_0)

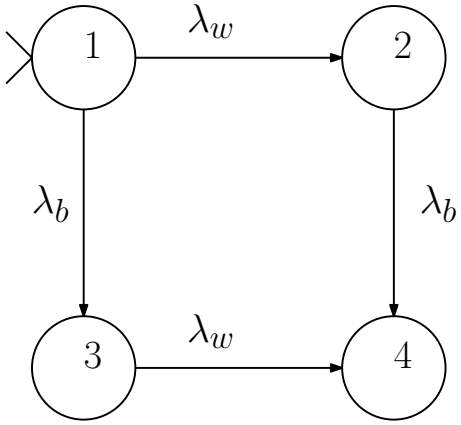


Figure 4.1 Transition diagram of Markov chain

4. Exercise

a) The machine starts out without biscuits. Both types of arrivals are enabled in the initial state (1 in Fig. 4.1). After the first arrival, only arrivals of the type that did not occur should be enabled (see states 2 and 3). Finally a state where the cookie can be produced is reached (state 4). [5]

b) The Markov chain has 4 states. We define the probability vector $\pi = [\pi_1, \pi_2, \pi_3, \pi_4]$. This evolves according to the equation:

$$\dot{\pi} = \pi Q$$

with the matrix Q given by:

$$Q = \begin{bmatrix} -\lambda_w - \lambda_b & \lambda_w & \lambda_b & 0 \\ 0 & -\lambda_b & 0 & \lambda_b \\ 0 & 0 & -\lambda_w & \lambda_w \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

c) Let 1 be the initial state (with probability 1), viz. $\pi(0) = [1,0,0,0]$. The average production time of a cookie is the average absorption time of state 4 in the

considered Markov chain.

$$\mathscr{L}(\dot{\pi}_4) = \pi(0)(sI_3 - Q_{1:3,1:3})^{-1} \begin{bmatrix} 0 \\ \lambda_b \\ \lambda_w \end{bmatrix}$$

$$= \frac{\lambda_b \lambda_w (\lambda_b + \lambda_w + 2s)}{(s+\lambda_b)(s+\lambda_w)(s+\lambda_b+\lambda_w)}.$$

The average absorption time T_a is then given by:

$$E[T_a] = \int_0^{+\infty} t\dot{\pi}_4(t)\,dt = -\frac{d}{ds}\mathscr{L}(\dot{\pi}_4)\Big|_{s=0} = \frac{\lambda_w^2 + \lambda_w\lambda_b + \lambda_b^2}{\lambda_b\lambda_w(\lambda_b+\lambda_w)}.$$

d) If cookies of arbitrary color are produced then the average production time is twice the average arrival time of biscuits, viz.

$$E[T_a] = \frac{2}{\lambda_b + \lambda_w}.$$

Notice that the arrival time of biscuits is also exponentially distributed, with rate $\lambda_w + \lambda_b$.